# Which lesson chapter should a Game Review moment link to?

The chapter that shows the same mistake

Chess Coach research lab

**Abstract**

The app's Game Review explains each key moment of a game, but it named no lesson, although the app has 74 study lessons. The daily review does name one: it reads what the engine's better move is for, with the board code, and offers the chapter whose recommended moves are read the same way (B). That matcher concentrates: one chapter takes 41.5% of the held-out moments it names. The lessons also play 114 wrong moves on purpose. The candidate (C) reads what his move got wrong on the board, from the move and the reply line the review's lesson uses, and for four kinds of mistake links the chapter whose wrong move is read the same way; otherwise it is B. On 223 held-out games, a blind judge, in both orders, preferred C at 96, tied 7 and preferred B at 10 of the 113 moments where both name a chapter and the chapters differ: (wins − losses) / judged = 0.76 (0.65 to 0.86). The controls held, coverage stayed, and the most named chapter's share fell to 36.9%. By the rule fixed before the run, C is kept.

## 1 The question

At a key moment where he went wrong, the review says what his move allowed and what the engine's move does instead. A link to the lesson that teaches the idea would turn that into study. The daily review already links one, by what the engine's move is for, and on his games one chapter, "Move the queen while taking the loose pawn" from "Choose a defense that changes the threat", is its pick at 41.5% of the held-out moments it names a chapter for, whatever his move got wrong. The lessons hold another source: the wrong moves they play on purpose, each in a chapter about that mistake. Can the board code read his mistake the way those chapters show it, and does a link to the chapter showing the same mistake fit the moment better?

## 2 What the app does

**What his move got wrong.** The Knowledge service's `mistakeOf` walks his move and the reply line the review's lesson shows after it (Table 1). A mate comes first: the line mates within three of their moves, with the mate's pattern when the board code names it. Otherwise the line must lose material. The reader takes their first capture in the first six plies, passing over their taking of the moved piece when that exchange does not lose for him, and names the first kind that fits that capture, or none.

**The lessons' wrong moves.** The scripts mark `counterexample` each wrong move the lessons play on purpose, and the study build reads them the same way, with the scene's moves after each as the line: 47 of the 114 are read as a kind, in 41 of the 493 chapters. The others lose nothing in the scene's line, or lose it in a way no kind names.

| Kind | What the board code sees | Lessons | Train | Held out | C links |
|---|---|---|---|---|---|
| `allows-mate` | the line mates within three of their moves | 16 | 15 | 40 | yes |
| `bad-capture` | his capture loses the exchange, and they take back on that square | 9 | 3 | 18 | yes |
| `unsafe-square` | their first move takes the moved piece and wins the exchange | 7 | 9 | 50 | no |
| `opens-line` | their first move captures along a line through the square his piece left | 4 | 5 | 7 | no |
| `deserted-guard` | the moved piece guarded a piece that now has no guard, and they take it within three plies | 5 | 10 | 25 | yes |
| `ignored-threat` | their first move takes a piece of his that was already loose | 4 | 22 | 94 | no |
| `allows-tactic` | a move of theirs in the line wins material with a tactic the board code names | 2 | 30 | 79 | yes |
| `loses-material` | the line loses two points or more, their capture within three plies | 0 | 16 | 82 | no |
| none | no kind fits, or the line loses no material | | 224 | 830 | |

Table 1: What his move got wrong, read from the move and the reply line the review's lesson shows, in the order the reader tries them at the capture it reads. *Lessons*: the lessons' wrong moves read as that kind. *Train*, *Held out*: his moments. *C links*: C names the chapter showing the same mistake for this kind; the others fall back to B.

**The match.** For his mistake, a chapter is offered when one of its wrong moves is read as the same kind, with the same idea (the mate's pattern, the tactic) or with no idea named, and its lesson is tagged with one of the mistake's concepts; a wrong move set in the opening counts only for a mistake in the opening. The same idea counts most, then the same phase, the lesson's topic, the same piece moved and the same piece lost. When no chapter shows the mistake and it has an idea, the chapters whose own recommended moves play that idea, in lessons that list it among their first two tags, are offered instead.

**The two matchers.** B is the daily review's: the first chapter offered for what the engine's move is for, else for what their reply is for. C is the chapter matched to his mistake when it is one of the four kinds the last column of Table 1 marks, else B. The code is the app's at commit `816c07c243`.

## 3 How it was measured

**The moments.** His rapid games in the app's archive, each reviewed the way the phone's Game Review reviews it. A moment is a key moment of the review where he moved, the move is graded an inaccuracy, mistake, miss or blunder, and the review writes a lesson for it.

**The split.** Train: one game in every eight up to 2026-09-16 (59 games, 334 moments), on which the change was designed. Held out: every game from 2026-09-17 to 2026-11-04 (223 games, 1,225 moments) except those of the three days read while it was designed (2026-10-31, 2026-11-02, 2026-11-03).

**The judge.** On the moments where B and C name different chapters (at most 60 on train and 120 held out), a model judge sees the moment as the review describes it (the position's FEN, his move and its grade, the engine's move, the reply line, the lesson's headline and its "difference" lines) and the two chapters, each as its lesson's title, its chapter's title and its narration, labelled A and B. It answers which chapter teaches what the moment needed, or that they are the same. Every pair is judged in both orders.

**The rule, fixed before the run.** C is kept when, held out, (wins − losses) / judged has its 95% interval above zero, from 10,000 bootstrap draws over games; C's coverage is at least B's minus 2 points; C's most named chapter takes no larger share than B's; on train, C's wins are not fewer than its losses; and the controls hold: 20 held-out moments with the same chapter on both sides are read as the same at least 16 times, and 20 with C's chapter against a chapter drawn from another lesson are won by C at least 14 times.

**Changed after the train round.** C first linked by the mistake for six kinds. On train (C 29 wins, 8 ties, 13 losses of 50), the judge favoured B on the `unsafe-square` and `opens-line` moments and C on every other kind (Table 3, first column), so those two now fall back to B. The change was made before any held-out verdict, and the rule was applied unchanged to the narrowed C.

## 4 The verdict

| | Train | Held out |
|---|---|---|
| Games (moments) | 59 (334) | 223 (1,225) |
| Moments where B and C name different chapters | 41 | 115 |
| judged (both name one) | 40 | 113 |
| **C wins / ties / losses** | **30 / 5 / 5** | **96 / 7 / 10** |
| **(wins − losses) / judged** (bar: interval above 0) | 0.63 (0.36 to 0.84) | **0.76 (0.65 to 0.86)** |
| **Coverage**, B; C (bar: C at least B − 2 points) | 86.8%; 87.1% | 87.4%; **87.6%** |
| **Most named chapter's share**, B; C (bar: C at most B) | 44.5%; 39.2% | 41.5%; **36.9%** |
| Distinct chapters (lessons), B; C | 18 (17); 28 (20) | 24 (21); 35 (25) |

Table 2: The rule's measures. A moment is a win for C when the judge preferred C in both orders, or in one with the other the same; a loss likewise for B. Intervals are 95% percentiles of 10,000 bootstrap draws over games. Coverage is the share of moments where the matcher names a chapter.

On held-out moments C wins 96 and loses 10 of 113 (Table 2), the controls hold (the same chapter read as the same 20 of 20 times; C beat the decoy 14 of 20 times), C names a chapter about as often as B, and it spreads the links over more chapters: 35 against 24. C is kept.

**Where the link comes from.** Held out, C names the chapter showing the same mistake at 120 moments, the chapter for what the engine's move is for at 623, the chapter for what their reply is for at 330, and none at 152. Beside the rule, each path was judged against a decoy chapter from another lesson (Table 4); the Game Review shows a link only from a path that reaches the bar.

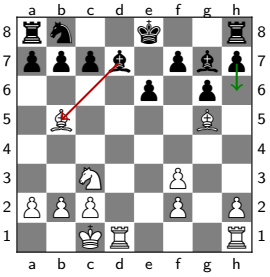
| His mistake | Train, six kinds | Train, four kinds | Held out, four kinds |
|---|---|---|---|
| `allows-mate` | 10 / 0 / 2 | 13 / 0 / 2 | 34 / 2 / 4 |
| `deserted-guard` | 8 / 1 / 1 | 8 / 1 / 1 | 23 / 1 / 0 |
| `bad-capture` | 2 / 1 / 0 | 2 / 1 / 0 | 15 / 2 / 1 |
| `unsafe-square` | 1 / 3 / 4 | – | – |
| `opens-line` | 1 / 0 / 4 | – | – |
| `allows-tactic` | 7 / 3 / 2 | 7 / 3 / 2 | 24 / 2 / 5 |
| All | 29 / 8 / 13 | 30 / 5 / 5 | 96 / 7 / 10 |

Table 3: C's wins / ties / losses by what his move got wrong. The first column is the train round of the candidate before it was narrowed; the kinds C no longer links by fall back to B and leave the judged pairs.

| C's chapter, from | Held-out moments | C wins | same | C loses | Shown |
|---|---|---|---|---|---|
| the same mistake (`mistake`) | 120 | 19 | 0 | 1 | yes |
| the engine's move (`fix`) | 623 | 15 | 3 | 2 | yes |
| their reply (`reply`) | 330 | 14 | 4 | 2 | yes |

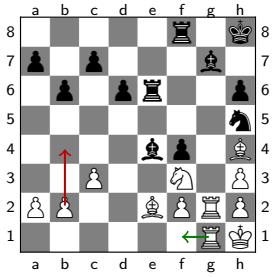
Table 4: Each way C names a chapter, against a chapter drawn at random from the other lessons: up to 20 held-out moments each, judged in both orders. A path is shown in the Game Review when C wins at least 14.

## 5 Three moments



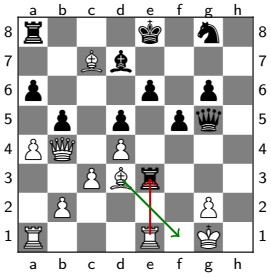
FEN `rn2k2r/pppb1pbp/4p1p1/1B4B1/8/2N2P2/PPP2P1P/2KR3R b kq - 3 12`

(a) Black to move. He played 12. . . ♗xb5 (`d7b5`, red); the engine preferred 12. . . h6 (`h7h6`, green). The line: 12. . . ♗xb5 13. ♖d8# (`d7b5 d1d8`). Read as `allows-mate` (`opera-mate`). B: "The checking queen is protected" from "Prove that the king has no escape". C: "A bishop move that abandons d2" from "Check what your move leaves behind".



FEN `5r1k/p1p3b1/1p1pr2p/7n/4bp1B/2P2N1P/PP2BPRP/6RK w - - 0 28`

(b) White to move. He played 28. b4 (`b2b4`); the engine preferred 28. ♖f1 (`g1f1`). The line: 28. b4 ♗b7 29. ♗d1 ♗xc3 (`b2b4 e4b7 e2d1 g7c3`). Read as `deserted-guard`. B: "The knight defends f7 and sets a trap" from "Improve with a tactic in reserve". C: "One bishop move removes both guards" from "Check what your move leaves behind".



FEN `r3k1n1/2Bb4/p3p1p1/1p1p1pq1/PQ1P4/2PBr3/1P4P1/R3R1K1 w q - 0 21`

(c) White to move. He played 21. ♖xe3 (`e1e3`); the engine preferred 21. ♗f1 (`d3f1`). The line: 21. ♖xe3 ♕xe3+ 22. ♔h1 ♕xd3 (`e1e3 g5e3 g1h1 e3d3`). Read as `allows-tactic` (`fork`). B: "Move the queen while taking the loose pawn" from "Choose a defense that changes the threat". C: "A knight attacks king, queen, and rook" from "Recognize tactics during an ordinary game".

Figure 1: Held-out moments where the judge preferred C in both orders, one for each of three kinds.

## 6 Limits

The judge is a language model reading the review's own words and the chapters' narration; it is told to take the review's facts as true and not to analyse the position, so it judges fit, not chess. Its fairness is checked only by the two controls, and the decoy control passed with C winning 14 of 20, where 14 were needed. By path (Table 4), the chapter showing the same mistake beat its decoy 19 of 20 times, the chapters B's two readings name 15 and 14 times: the links that come from B are the weaker part of C. The moments are one player's, at his level. The mistake reader names a kind from one reply line, the one the review shows; a move can be wrong in more ways than that line shows. A kind with few wrong moves in the lessons links few chapters, so C's links stay concentrated where the lessons show only one example.

## 7 What the app took

The Game Review's lesson panel, at his key moments, ends with the chapter C names, from the paths Table 4 marks as shown, as a link to watch or read the lesson with a line on why it fits. The study build publishes the chapters with what each one's moves are for and get wrong, so the review links one without loading every lesson.